Soft Suave
A KiwiTech Affiliate Company

CASE STUDY · DRIVER SAFETY · BEHAVIORAL AI

DRIVER SAFETY & TRAINING · CLOUD AI PLATFORM BUILD

Neurora
Behavioral Intelligence

A Platform for Driver Attention, Stress and Safety

Fusing synchronized driving video, eye-gaze and heart-rate data into automated behavioral scores — showing researchers, fleets and driving schools not just what a driver did, but where they looked, what they missed and when stress took over.

INDUSTRY
Driver safety, training & behavioral research

SCOPE
Mobile capture, AWS AI pipeline, analytics dashboard

ENGAGEMENT
PoC-first end-to-end build · 5 months

01 — ABOUT THE CLIENT

Where driver safety is still judged by eye

Neurora Labs is building a Behavioral Intelligence Platform for organizations that need to understand how people actually drive. Its users are driving schools, fleet safety and risk teams, academic and commercial driving-behavior researchers, and automotive safety engineers. Each of them depends on evidence from real driving sessions — video, gaze, heart rate — and each has been forced to rely on an instructor's memory and judgement to interpret it.

02 — THE CHALLENGE

Four gaps between the drive and the insight

Manual, after-the-fact review

Instructors scored sessions from observation and post-drive recall, so feedback arrived late and its quality varied with whoever happened to be watching.

No objective attention data

Nothing recorded where a driver actually looked or what they missed. Hazard awareness was inferred from behavior, never measured directly.

Sensors that never talked

Existing tools captured video without gaze, or tracked eyes without driving context. No product fused video, gaze and physiology on one timeline.

Subjective, inconsistent scoring

Safety judgements rested on individual opinion, making results hard to compare across drivers, sessions or organizations, and impossible to benchmark at scale.

03 — THE SOLUTION

One capture-to-insight pipeline, three connected surfaces

Soft Suave is delivering an end-to-end system for Neurora Labs: Neurora Capture records synchronized video, gaze and heart-rate data; an AWS backend runs SQS-triggered ECS workers for computer vision and scoring; and Neurora Insight, the Neurora Self View driver app and an admin panel turn the output into decisions.

01 Neurora Capture

MOBILE · MULTI-SENSOR

A mobile recording app that captures camera video, eye-gaze coordinates and pupil size from wearable tracking glasses, and heart-rate signals from a wearable sensor — all timestamped and synchronized on-device.

Neurora Capture	
Video	00:42:18
Gaze + pupil	synced
Heart rate	78 bpm

02 AI Processing Pipeline

AWS · CV + SCORING

Uploaded sessions trigger cloud workers that detect and track objects across frames, map gaze to dynamic Areas of Interest, and compute fixation duration, PERCLOS, time-to-first-fixation and gaze entropy.

AI Processing Pipeline	
Objects tracked	1,284
AOI mapping	complete
PERCLOS	0.12

03 Insight, Self View & Admin

DASHBOARD · SELF-VIEW

A React dashboard replays the drive with gaze overlay, heatmaps, scan paths and bounding boxes alongside safety, focus, stress and risk scores — plus the Self View driver app and an admin panel for user and session control.

Insight, Self View & Admin	
Safety score	82 / 100
Focus	high · Stress: low
Risk events	3

Raw sensor streams in, behavioral scores and plain-language safety insight out — automated end to end, with human-adjustable Areas of Interest.

04 — KEY FEATURES

Twelve capabilities, one platform

- Synchronized Video, Gaze & Heart-Rate Capture
- AI Object Detection & Frame Tracking
- Dynamic Area-of-Interest Gaze Mapping
- Fixation, PERCLOS, TTFF & Gaze Entropy
- Automated Safety, Focus & Risk Scoring
- Stress & Cognitive Load Modeling
- Attention Heatmaps & Scan-Path Replay
- AI-Generated Plain-Language Session Insights
- Hybrid Manual AOI Adjustment
- Neurora Self View Driver App
- Report Export & Session Sharing
- Secure AWS Storage & Admin Panel

05 — RESULTS & IMPACT

A delivery plan built to de-risk the platform

5 months
End-to-end delivery from kick-off to complete application, across a milestone-driven release plan

2 streams
AI and application development run in parallel — 12 weeks and 10 weeks — to cut time-to-market

2–3 weeks
Milestone 0 proof of concept, delivered at no cost with Soft Suave bearing PoC hosting costs

90 days
Post-deployment support by the original project team after go-live

WHY SOFT SUAVE

Multi-sensor AI, delivered on a proven cadence

This is not a dashboard project — it is sensor fusion, computer vision, gaze mapping and behavioral modeling delivered as one production platform. Soft Suave brings 13+ years of engineering depth and a milestone-driven agile cadence: a no-cost proof of concept first, then parallel AI and application streams tracked in Jira and Confluence with full client visibility. Every milestone ships to a client-owned repository, backed by layered functional, integration and regression testing and a 90-day post-deployment support period.

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Let's build what's next.